VEHICLE SERVICE CENTER MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment of requirements for the award of the course

CGB1221-DATABASE MANAGEMENT SYSTEMS

In

COMPUTER SCIENCE AND ENGINEERING

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE, New Delhi)

SAMAYAPURAM – 621 112

JUNE- 2025

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY (AUTONOMOUS)

SAMAYAPURAM – 621 112

BONAFIDE CERTIFICATE

Certified that this project report on **"VEHICLE SERVICE CENTER MANAGEMENT SYSTEM "** is the bonafide work of **PRADHIKSHA R (2303811710422117)** who carried out the project work during the academic year 2024 - 2025 under my supervision.

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INTERNAL EXAMINER EXTERNAL EXAMINER

DECLARATION

I declare that the project report on**"VEHICLE SERVICE CENTER MANAGEMENT SYSTEM "** is the result of original work done by us and best of our knowledge, similar work has not been submitted to **"ANNA UNIVERSITY CHENNAI"** for the requirement of Degree of BACHELOR OF ENGINEERING.This project report is submitted on the partial fulfilment of the requirement of the completion of the course **CGB122 – DATABASE MANAGEMENT SYSTEM..**

.

Signature

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ACKNOWLEDGEMENT

It is with great pride that I express our gratitude and in-debt to our institution "**K.Ramakrishnan College of Technology (Autonomous)**", for providing us with the opportunity to do this project.

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VISION OF THE INSTITUTION

To serve the society by offering top-notch technical education on par with global standards

MISSION OF THE INSTITUTION

Be a center of excellence for technical education in emerging technologies by exceeding the needs of the industry and society.

Be an institute with world class research facilities

Be an institute nurturing talent and enhancing the competency of students to transform them as all-round personality respecting moral and ethical values

VISION OF DEPARTMENT

To be a center of eminence in creating competent software professionals with research .and innovative skills

MISSION OF DEPARTMENT

M1: Industry Specific: To nurture students in working with various hardware and software platforms inclined with the best practices of industry.

M2: Research: To prepare students for research-oriented activities.

M3: Society: To empower students with the required skills to solve complex technological problems of society.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1: Domain Knowledge .1

To produce graduates who have strong foundation of knowledge and skills in the field .of Computer Science and Engineering

PEO2: Employability Skills and Research .2

To produce graduates who are employable in industries/public sector/research .organizations or work as an entrepreneur

PEO 3: Ethics and Values

To develop leadership skills and ethically collaborate with society to tackle real-world challenges.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1: Domain Knowledge

To analyze, design and develop computing solutions by applying foundational concepts of Computer Science and Engineering.

PSO 2: Quality Software

To apply software engineering principles and practices for developing quality software for scientific and business applications

PSO 3: Innovation Ideas

To adapt to emerging Information and Communication Technologies (ICT) to innovate ideas and solutions to existing/novel problems

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations

The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice

Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project manaGHJKL;'gement and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The Vehicle Service Center Management System is a project that helps manage the daily operations of a vehicle service center using a Database Management System (DBMS). In a typical service center, it can be difficult to handle customer details, vehicle records, service appointments, bills, and mechanic assignments manually. This system is designed to make those tasks easier, faster, and accurate.

This project stores all important data in a relational database, such as customer information, vehicle details, service history, available spare parts, and billing records. Using DBMS features like tables, relationships, and SQL queries, the system can add, update, search, and delete records efficiently. It also reduces chances of error & avoids repeated the same data and its organized user-friendly.

The system has different roles like admin, staff, and mechanics, each with different access levels. For example, staff can book service appointments and issue bills, while mechanics can update the service status of vehicles. The system also generates reports like service history, parts used, and total income. It helps to improve customer satisfaction, and keeps all records in access and in safe file.

ABSTRACT WITH POs AND PSOs MAPPING

CO 5 : BUILD JAVA APPLICATIONS FOR SOLVING REAL-TIME PROBLEMS.

ABSTRACT	POs MAPPED	PSOs MAPPED
The Vehicle Service Center Management System is a database project designed to manage the operations of a vehicle service center efficiently. It helps store and organize customer details, vehicle information, service records, and billing using a relational database. The system reduces manual work, avoids data errors, and makes it easy to track services and generate reports. Overall, it improves the speed and accuracy of service center tasks using DBMS concepts.	PO1 -3 PO2 -3 PO3 -3 PO4 -3 PO5 -3 PO6 -3 PO7 -3 PO8 -3 PO9 -3 PO10 -3 PO11-3 PO12 -3	PSO1 -3 PSO2 -3 PSO3 -3

Note: 1- Low, 2-Medium, 3- High

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CHAPTER 1

INTRODUCTION

1.1Objective

The objective of the Vehicle Service Center Management System is to design and implement a database-driven application that streamlines and automates the operations of a vehicle service center. This system aims to manage the interactions between customers, mechanics, service requests, and service records efficiently. It reduces manual paperwork, ensures data consistency, and provides timely tracking of vehicle maintenance activities. The system enables user authentication, real-time updates, and reporting functionalities that benefit both the management and the customers by improving service delivery and communication.

1.2Overview

The Vehicle Service Center Management System is a centralized platform developed using database management principles to manage and coordinate various activities of a vehicle repair shop or service center. It facilitates efficient handling of customer records, service histories, employee (mechanic) assignments, and billing.

1.2.1. Customer Management

- Stores customer personal information, contact details, and vehicle information.
- It keeps a history of previous services and repairs.

1.2.2. Service Management

- Tracks service requests, assigns them to mechanics, and logs service completion.
- Maintains service history records for future reference.

1.2.3. Mechanic Management

- Contains mechanic profiles, specialties, availability, and assigned tasks.
- Monitors workload distribution and performance.

1.2.4. Appointment Scheduling:

- Allows customers to schedule appointments and service requests online or via the front desk.
- Sends reminders and updates.

1.3 SQL and Database Concepts

- Entity-Relationship (ER) Modeling:

  ER modeling helps design the database by identifying entities like Customer, Vehicle, and Service, and the relationships between them. This aids in building a structured, normalized schema before implementation.

- Normalization:

  Normalization organizes the database into tables to reduce data redundancy and improve integrity. The system follows 1NF, 2NF, and 3NF to ensure efficiency and eliminate anomalies.

- SQL DDL (Data Definition Language):

  DDL commands like CREATE, ALTER, and DROP are used to define and modify database structure. Tables for customers, services, and vehicles are created using these commands.

- SQL DML (Data Manipulation Language):

  DML commands (INSERT, UPDATE, DELETE, SELECT) manage the data inside tables. These are used to add service records, update statuses, and retrieve service histories.

CHAPTER 2
PROJECT METHODOLOGY

2.1 Proposed Work

The proposed system is a web-based or desktop-based application integrated with a backend SQL database that centralizes all service center operations. It focuses on real-time data processing, improves data accuracy, and ensures secure user interactions. The system provides interfaces for different stakeholders—customers, mechanics, and administrators—allowing for transparency and smooth operations.

Key Features of the Proposed System

1. **User Authentication:**

- Role-based access control for Admin, Customer, and Mechanic.
- Secure login with password encryption.

2.**Service Request Handling:**

- Customers can request services and track their status.
- Admin assigns service tasks to mechanics.

3. **Mechanic Dashboard:**

- View assigned tasks, update job status, and add notes.
- View schedules and workload.

4. **Vehicle and Service Record Keeping:**

- Maintain detailed vehicle information and service history.
- Generate reports for analytics.

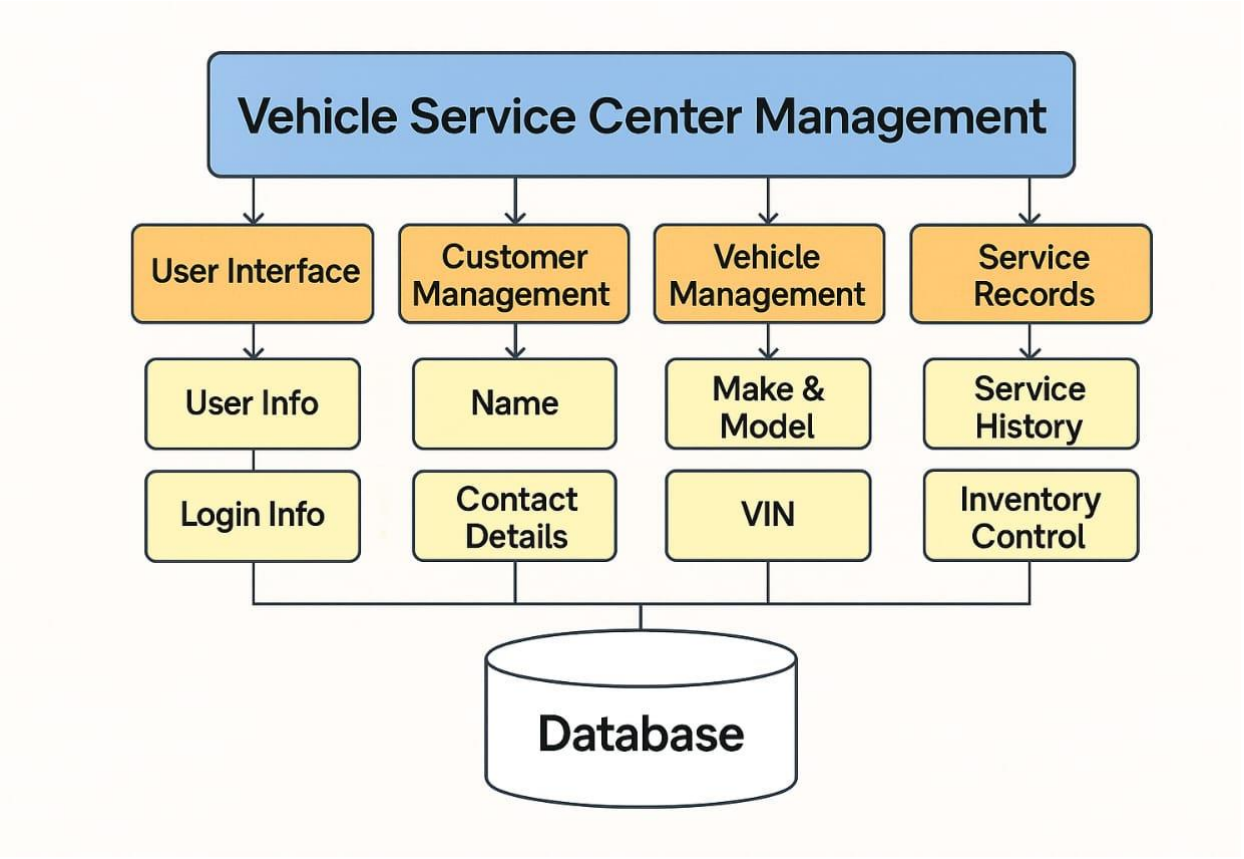
5. **Automated Billing System:**

- Calculate service costs based on tasks and parts.
- Generate digital invoices and record payments.

6.**Notifications and Alerts:**

- Send service reminders, completion alerts, and appointment confirmations via SMS or email.

2.2 Block Diagram



CHAPTER 3
MODULE DESCRIPTION

3.1. User Management Module

This module handles login, logout, and session management for all users. It ensures secure access to the system through username and password authentication.

Users are assigned roles such as Admin, Staff, and Customer, each with different access levels. Admins have full control over the system, while staff can manage services and inventory. Customers can log in to view their vehicle and service history.

Role-based access ensures users can only perform tasks relevant to their role. This module improves data security and system integrity. It helps prevent unauthorized access to sensitive business data. Password recovery and account management features can also be included. Role-based access ensures users can only perform tasks relevant to their role.

3.2 Customer Management

This module allows the system to add, update, delete, and search customer records . Each customer's profile includes essential data like name, contact info, email, and address .The system links each customer with their registered vehicle.. Search functionality enables quick access to customer details. Customer records support follow-ups, reminders, and feedback collection. This module ensures organized and efficient customer handling. Admins and staff can filter data based on criteria like vehicle or service history. It enhances customer relationship management and communication. This module plays a key role in tracking service history and communication with customers. Well-maintained customer records improve service personalization and customer satisfaction . In short, it acts as the central hub for managing client relationships.

3.3 Vehicle Management Module

This module tracks vehicle-related information and their linkage to customers. Users can add new vehicles, update existing records, or remove outdated ones. Vehicle fields include model, make, year, registration number, VIN, and owner. Each vehicle is associated with a specific customer in the system. Service history is tied to each vehicle, ensuring a detailed maintenance log. Mechanics can easily view previous issues or repairs done on a vehicle. The module helps reduce errors by storing all vehicle data in one place. It supports better planning for services and part replacements. Overall, it ensures smooth management of fleet and individual vehicles.

3.4 Service management Module

This module logs and tracks all services performed on vehicles. Users can create new service records including service type, remarks, and assigned staff . Service types include maintenance, repair, diagnostics, and more. Each service record is linked to a specific customer and vehicle. The module tracks service status like Pending, In Progress, or Completed. Mechanics can add notes or diagnoses to each record for future reference. Admins can monitor workload and staff performance. Customers can receive updates about the status of their service. It helps generate service reports that are useful for billing and performance reviews. Admin can monitor technician performance and service turnaround time. This module improves workflow efficiency and customer satisfaction by keeping services well-organized. It supports assigning services to specific mechanics or technicians. Service status is tracked using labels like Pending, In Progress, or Completed. Admins can monitor workload and staff performance.

3.5 Spareparts inventory Module

This module manages all spare parts available at the service center. Each part entry includes part name, number, manufacturer, cost, quantity, and date added . Admins can add, update, or remove spare parts from the system. It tracks stock levels in real-time, warning when parts are low or out of stock. Spare parts can be assigned to specific service jobs, linking costs automatically. The module helps in avoiding overstocking or stockouts, improving inventory control. Inventory reports can be generated for purchasing decisions and audits. Admins can view usage history to determine which parts are in high demand. It ensures the availability of required parts, improving service delivery speed. The spare parts inventory system supports financial transparency and resource planning. The system deducts inventory when parts are used in a service job. Stock alerts notify staff when items are running low.

CHAPTER 4
CONCLUSION AND FUTURE ENHANCEMENT

4.1. CONCLUSION

The Vehicle Service Center Management System is a powerful solution that addresses the critical needs of automotive service centers through the use of a centralized, database-driven application. By integrating core service processes like customer registration, service request management, mechanic task handling, billing, and vehicle history tracking, the system significantly enhances the overall efficiency and reliability of service delivery. It reduces human error, minimizes paperwork, and improves transparency between the service provider and the customer.

The project demonstrates how database concepts and well-structured modules can be effectively combined to solve real-world problems and modernize traditional service operations.

4.2. FUTURE ENHANCEMENTS

To make the system more robust, scalable, and user-friendly, several enhancements can be planned for future versions:

1. Mobile Application Integration:
   A dedicated Android/iOS mobile app would allow customers and mechanics to access the system on the go, improving usability and engagement.
2. QR Code Vehicle Profiles:
   Each vehicle can be tagged with a QR code that links to its digital service history, simplifying check-ins and record retrieval during service appointments.
3. Predictive Maintenance Using AI/ML:
   Incorporating machine learning algorithms can enable the system to predict

potential issues based on service data, offering proactive maintenance alerts.

4. Inventory and Spare Parts Management:
   Integrating inventory control would allow real-time tracking of spare parts, auto-reordering, and better supply chain management.
5. Multi-Branch Support:
   The system can be extended to support multi-location service centers with centralized reporting and management.
6. Integration with Vehicle Telematics:
   For advanced vehicles, integrating with telematics systems can allow real-time diagnostics and service alerts based on actual vehicle conditions.

APPENDIX A

1. db.php – Database Connection

```
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "vehicle_service";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
    }
```

2. login.php-Login Page

```
    <link rel="stylesheet" href="style.css">
    <?php
    session_start();
    $error = '';
    if ($_SERVER["REQUEST_METHOD"] == "POST") {
        $username = $_POST['username'] ?? '';
        $password = $_POST['password'] ?? '';
        if ($username === 'admin' && $password === 'admin123') {
            $_SESSION['username'] = $username;
            header("Location: dashboard.php");
            exit();
        } else {
```

```
$error = "Invalid username or password.";
    }
}
?>
<!DOCTYPE html>
<html>
<head>
    <title>Login - Vehicle Service Center</title>
    <style>
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background-color: #f2fefc;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .login-box {
            background-color: #ffffff;
            padding: 40px 30px;
            border-radius: 15px;
            box-shadow: 0 8px 20px rgba(0,0,0,0.1);
            width: 350px;
            text-align: center;
```

```
}h1 {
    margin-bottom: 25px;
    color: #009688;
    font-weight: 700;
    font-size: 28px;
    letter-spacing: 1px;
}
input[type="text"],
input[type="password"] {
    width: 100%;
    padding: 14px 12px;
    margin: 15px 0;
    border: 1.5px solid #ccc;
    border-radius: 8px;
    font-size: 16px;
    transition: border-color 0.3s ease;
}
input[type="text"]:focus,
input[type="password"]:focus {
    outline: none;
    border-color: #009688;
    box-shadow: 0 0 6px #009688;
}
input[type="submit"] {
    width: 100%;
```

```
            padding: 14px 0;
            background-color: #009688;
            color: white;
            border: none;
            border-radius: 8px;
            font-weight: 600;
            font-size: 18px;
            cursor: pointer;
            margin-top: 20px;
            transition: background-color 0.3s ease;
        }
        input[type="submit"]:hover {
            background-color: #00796b;
        }
        .error {
            color: #e74c3c;
            margin-top: 15px;
            font-weight: 600;
        }
    </style>
</head>
<body>
<div class="login-box">
    <h1>Vehicle Service Center</h1>
<?php if ($error): ?>
```

```
        <div class="error"><?php echo htmlspecialchars($error); ?></div>
    <?php endif; ?>
    <form method="post" action="login.php" autocomplete="off">
        <input type="text" name="username" placeholder="Username" required
/>
        <input type="password" name="password" placeholder="Password"
required />
        <input type="submit" value="Login" />
    </form>
</div>
</body>
</html>
```

3. **dashboard.php**

```
<link rel="stylesheet" href="style.css">
<?php
session_start();
if (!isset($_SESSION['username'])) {
    header("Location: login.php");
    exit();
}
?>
<!DOCTYPE html>
<html lang="en"><head>
```

```
<meta charset="UTF-8">
<title>Dashboard - Vehicle Service System</title>
<style>
    * {
        box-sizing: border-box;
    }
    body {
        margin: 0;
        font-family: 'Segoe UI', sans-serif;
        background-color: #f2fefc;
    }
    header {
        background-color: #009688;
        color: white;
        padding: 20px 40px;
        display: flex;
        justify-content: space-between;
        align-items: center;
    }
    header h1 {
        margin: 0;
        font-size: 24px;
    }
    .logout-btn {
        background-color: #e74c3c;
```

```
    color: white;
    padding: 10px 18px;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-weight: bold;
}
.logout-btn:hover {
    background-color: #c0392b;
}
.container {
    padding: 40px;
}
.grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
    gap: 30px;
}
.card {
    background-color: white;
    padding: 25px;
    border-radius: 12px;
    box-shadow: 0 4px 12px rgba(0,0,0,0.08);
    text-align: center;
    transition: transform 0.2s, box-shadow 0.2s;
```

```
        }
        .card:hover {
            transform: translateY(-5px);
            box-shadow: 0 6px 16px rgba(0,0,0,0.12);
        }
        .card a {
            text-decoration: none;
            color: #009688;
            font-weight: bold;
            font-size: 18px;
        }
        .card a:hover {
            color: #00796b;
        }
        footer {
            text-align: center;
            padding: 20px;
            font-size: 14px;
            color: #666;
        }
    </style>
</head>
<body>
<header>
    <h1>Vehicle Service Center - Dashboard</h1>
```

```
    <form method="post" action="logout.php">
        <button class="logout-btn">Logout</button>
    </form>
</header>
<div class="container">
    <div class="grid">
        <div class="card"><a href="add_customer.php">➕ Add Customer</a></div>
        <div class="card"><a href="view_customers.php">👥 View Customers</a></div>
        <div class="card"><a href="add_spareparts.php">⚙️ Add Spare Parts</a></div>
        <div class="card"><a href="view_spareparts.php">📦 View Spare Parts</a></div>
        <div class="card"><a href="add_service.php">🛠️ Add Service Record</a></div>
        <div class="card"><a href="view_services.php">📋 View Service Records</a></div>
    </div>
</div>
<footer>
    &copy; <?php echo date("Y"); ?> Vehicle Service Center. All rights reserved.
</footer>
```

```
</body>
</html>
```

4. logout.php

```
<?php
session_start();
session_destroy();
header("Location: login.php");
exit();
?>
```

5. add_customer.php

```
<?php
session_start();
ini_set('display_errors', 1);
error_reporting(E_ALL);
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "vehicle_service";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
 die("Connection failed: " . $conn->connect_error);
  }
```

```
$success = "";
$error = "";
// Handle form submission
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $name = $_POST['name'] ?? '';
    $phone = $_POST['phone'] ?? '';
    $service_date = $_POST['service_date'] ?? '';
    $service_details = $_POST['service_details'] ?? '';
    $status = 'Processing';
    $stmt = $conn->prepare("INSERT INTO customers (name, phone,
service_date, service_details, status) VALUES (?, ?, ?, ?, ?)");
    $stmt->bind_param("sssss", $name, $phone, $service_date,
$service_details, $status);
    if ($stmt->execute()) {
        $success = "Customer added successfully!";
    } else {
        $error = "Error: " . $stmt->error;
    }
    $stmt->close();
}
?>
<!DOCTYPE html>
<html>
<head>
    <title>Add Customer</title>
```

```
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
    <h2>Add Customer</h2>
    <?php if ($success): ?>
        <p class="success"><?php echo $success; ?></p>
    <?php endif; ?>
    <?php if ($error): ?>
        <p class="error"><?php echo $error; ?></p>
    <?php endif; ?>
    <form method="post" action="">
        <label for="name">Customer Name:</label>
        <input type="text" id="name" name="name" required>
        <label for="phone">Phone Number:</label>
        <input type="text" id="phone" name="phone" required>
        <label for="service_date">Service Date:</label>
        <input type="date" id="service_date" name="service_date"
required>
        <label for="service_details">Service Details:</label>
        <textarea id="service_details" name="service_details" rows="4"
required></textarea>
        <button type="submit">Add Customer</button>
    </form>
</div>
```

```
</body>
</html>
```

6. **add_spareparts.php**

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "vehicle_service";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) die("Connection failed: " . $conn->connect_error);
$success = "";
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $part_name = $_POST["part_name"];
    $part_number = $_POST["part_number"];
    $manufacturer = $_POST["manufacturer"];
    $price = $_POST["price"];
    $quantity = $_POST["quantity"];
    $added_date = $_POST["added_date"];
    $stmt = $conn->prepare("INSERT INTO spare_parts (part_name,
part_number, manufacturer, price, quantity, added_date) VALUES (?, ?, ?, ?,
?, ?)");
    $stmt->bind_param("sssdis", $part_name, $part_number, $manufacturer,
$price, $quantity, $added_date);
    if ($stmt->execute()) {
        $success = "Spare part added successfully.";
```

```
    } else {
        $success = "Error: " . $conn->error;
    }
    $stmt->close();
}
$conn->close();
?>
<!DOCTYPE html>
<html>
<head>
    <title>Add Spare Part</title>
    <link rel="stylesheet" href="teal_style.css">
</head>
<body>
<div class="form-box">
    <h2>Add Spare Part</h2>
    <?php if ($success): ?>
        <div class="success"><?php echo $success; ?></div>
    <?php endif; ?>
    <form method="post" action="">
        <label>Part Name:</label>
        <input type="text" name="part_name" required>
        <label>Part Number:</label>
        <input type="text" name="part_number" required>
        <label>Manufacturer:</label>
```

```
        <input type="text" name="manufacturer" required>
        <label>Price:</label>
        <input type="number" name="price" step="0.01" required>
        <label>Quantity:</label>
        <input type="number" name="quantity" required>
        <label>Added Date:</label>
        <input type="date" name="added_date" required
        <div class="actions">
            <input type="submit" value="Add Spare Part">
            <a href="dashboard.php" class="back-btn">← Back to
Dashboard</a>
        </div>
    </form>
</div>
</body>
</html>
```

7. **teal_style.css**

```
body {
    font-family: Arial, sans-serif;
    background-color: #f1f1f1;
    margin: 0;
    padding: 40px;
}
.form-box {
    background-color: #ffffff;
```

```
    padding: 25px 30px;
    border-radius: 10px;
    box-shadow: 0 0 15px rgba(0, 150, 136, 0.2);
    max-width: 500px;
    margin: auto;
}
h2 {
    color: #009688;
    text-align: center;
    margin-bottom: 20px;
}
label {
    display: block;
    margin-top: 10px;
    font-weight: bold;
    color: #333;
}
input[type="text"],
input[type="number"],
input[type="date"],
textarea {
    width: 100%;
    padding: 8px;
    margin-top: 6px;
    margin-bottom: 16px;
```

```
    border: 1px solid #ccc;
    border-radius: 5px;
    box-sizing: border-box;
}
input[type="submit"] {
    background-color: #009688;
    color: white;
    padding: 10px 18px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    font-weight: bold;
    width: 100%;
    margin-top: 10px;
}
input[type="submit"]:hover {
    background-color: #00796b;
}
.success {
    background-color: #dff0d8;
    color: #3c763d;
    padding: 10px;
    margin-bottom: 15px;
    border-radius: 4px;
    text-align: center;
```

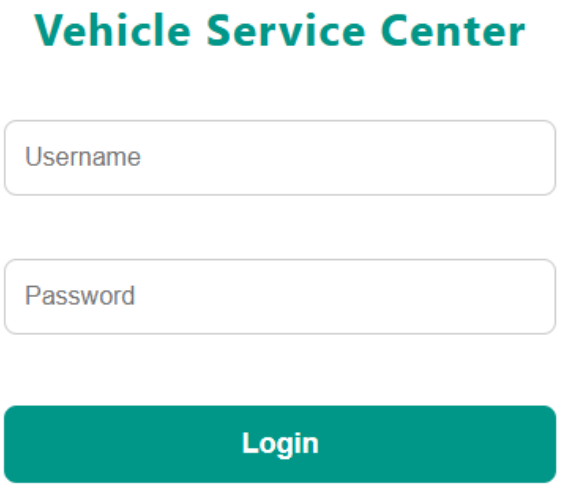
```
}
.actions {
    margin-top: 20px;
    text-align: center;
}
.back-btn {
    display: inline-block;
    margin-top: 10px;
```

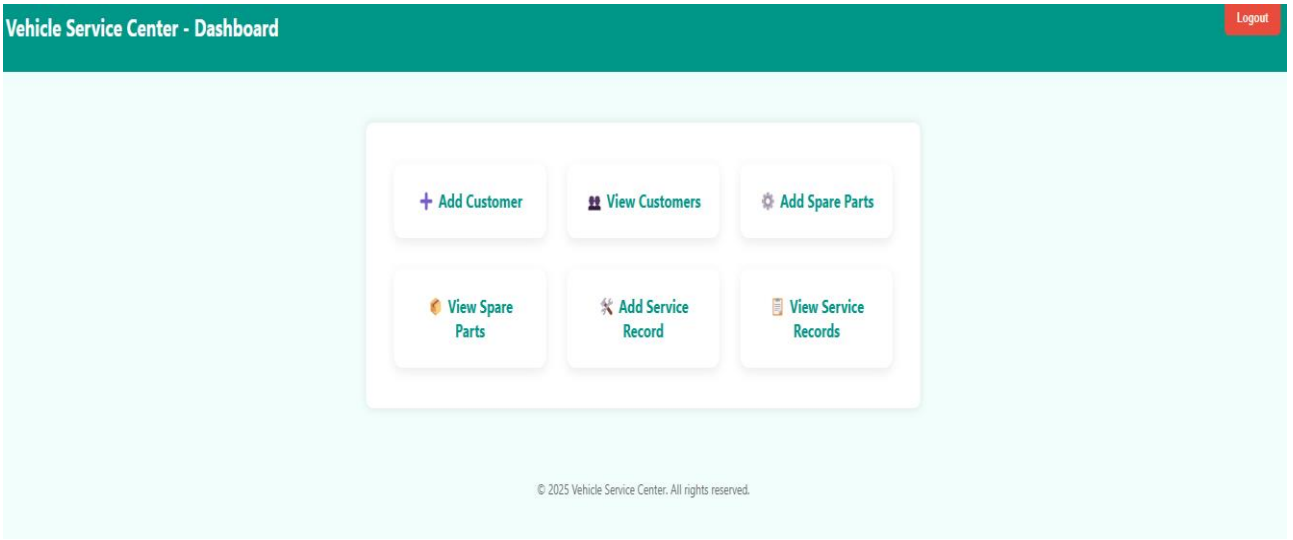
12 .SQL

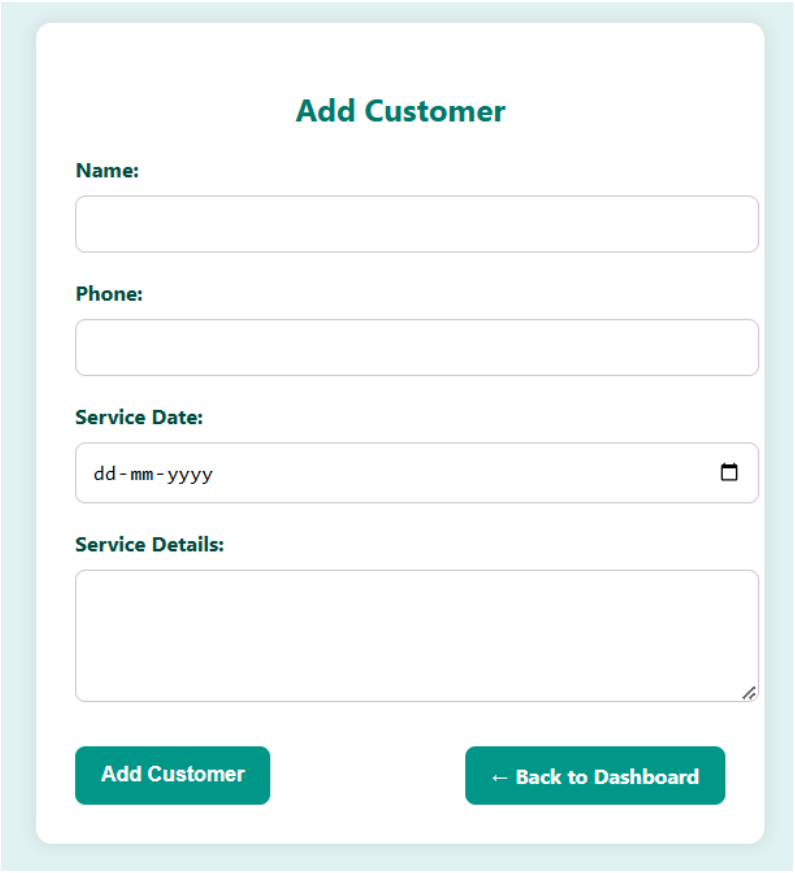
```
CREATE TABLE spare_parts (
id INT AUTO_INCREMENT PRIMARY KEY,
part_name VARCHAR(100) NOT NULL,
part_number VARCHAR(50) NOT NULL,
manufacturer VARCHAR(100),
price DECIMAL(10,2),
quantity INT,
added_date DATETIME DEFAULT CURRENT_TIMESTAMP
);
CREATE TABLE services (
id INT AUTO_INCREMENT PRIMARY KEY,
customer_id INT NOT NULL,
service_type VARCHAR(100) NOT NULL,
remarks TEXT,
service_date DATE NOT NULL
);
```

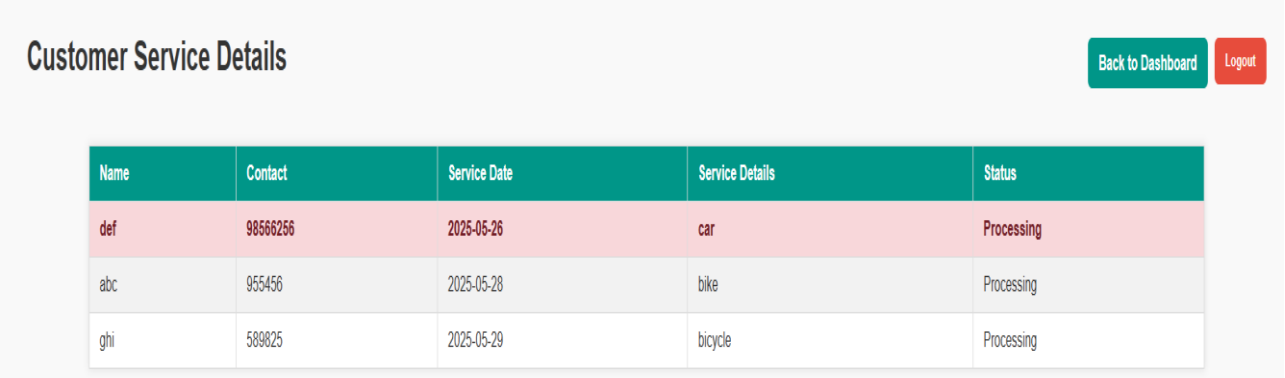
```
CREATE TABLE customers (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    phone VARCHAR(20),
    email VARCHAR(100),
    vehicle_model VARCHAR(100),
    service_date DATE,
    service_details TEXT,
    status VARCHAR(50)
);
```

APPENDIX B


Vehicle Service Center
Username
Password
Login


Vehicle Service Center - Dashboard
Logout
Add Customer
View Customers
Add Spare Parts
View Spare Parts
Add Service Record
View Service Records
© 2025 Vehicle Service Center. All rights reserved.





Name	Contact	Service Date	Service Details	Status
def	98566256	2025-05-26	car	Processing
abc	955456	2025-05-28	bike	Processing
ghi	589825	2025-05-29	bicycle	Processing

Add Spare Part

Part Name:

Part Number:

Manufacturer:

Price:

Quantity:

Added Date:

dd - mm - yyyy

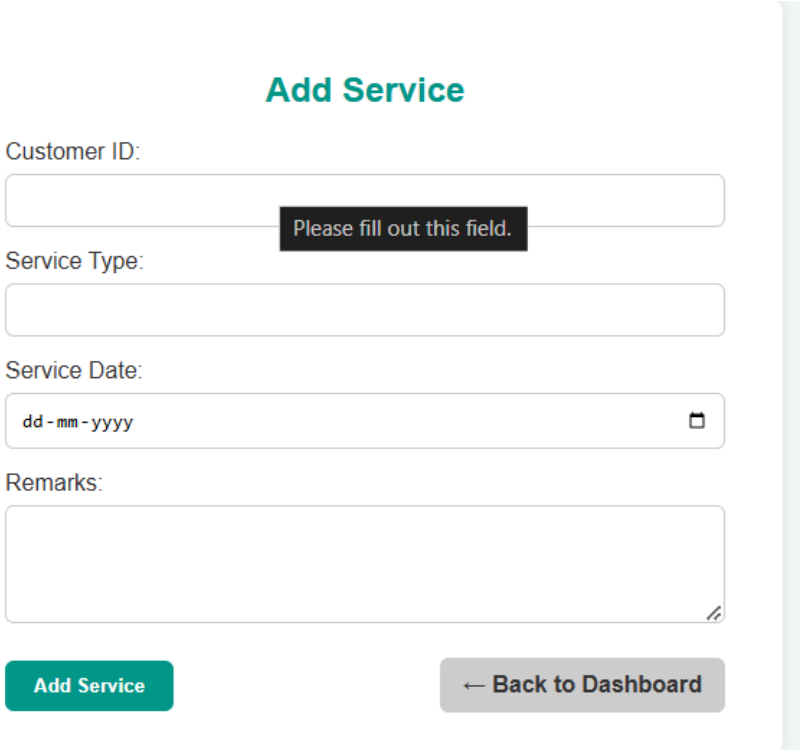
Add Spare Part

← Back to Dashboard

Spare Parts List

Back to Dashboard Logout

ID	Part Name	Part Number	Manufacturer	Price	Quantity	Added Date
1	brakepads	34	xxy	350.00	50	2025-05-30 00:00:00



Back to Dashboard | Logout

ID	Customer ID	Customer	Service Date	Service Type	Remarks
1	23	Unknown	2025-05-30	oil change	customer requested sysnthetic oil

REFERENCES:

- https://www.researchgate.net/publication/363700029_THE_VEHICLE_SERVICE_MANAGEMENT_SYSTEM

- chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.jetir.org/papers/JETIR2405968.pdf

- https://github.com/bhaveshgadag/vehicle_management_python

- https://www.sanfoundry.com/best-reference-books-service-engineering-management/#google_vignette